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Environmental Impact Mitigation Through Innovative Waste Gas Incineration

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Abstract

This paper evaluates an incinerator system that integrates advanced waste gas incineration with renewable energy aiming to optimize waste management reduce emissions. The study explores this technology's capacity to address the specific needs of the petroleum industry, including handling hazardous waste and improving sustainability in regions like the UAE & KSA. The development involved a comprehensive approach combining computational modelling (CFD), prototype testing, and field trials, utilizing CFD simulations for combustion process refinement. A prototype was deployed for the first time in the region in Abu Dhabi field, in close proximity to a populated area, where elimination of visual flare, emissions and odours was required, while incorporating solar energy for ignition. Performance metrics such as emission levels, energy efficiency, and adaptability to diverse waste streams were rigorously tested to ensure compliance with stringent environmental and industry standards. A life cycle assessment (LCA) was also conducted to evaluate the system's holistic environmental impact. The hybrid incinerator demonstrated remarkable results, achieving over 99.9% total hydrocarbon (including Volatile Organic Compounds (VOCs) and BTEX compounds) destruction efficiency, effectively eliminating free carbon particles and significantly reducing CO emissions to below 10 ppm. The system proved capable of safely handling diverse hazardous gas streams, including H_2S . Economically, it offers a high return on investment (ROI) due to notably lower construction costs compared to conventional enclosed flare systems that can achieve the same results. These findings suggest the incinerator can significantly reduce the petroleum industry's environmental footprint, promote energy recovery, and align with circular economic principles. The system meets and exceeds regulations from the United States Environmental Protection Agency (EPA), Alberta Energy Regulator (AER), Saskatchewan MOE regulations, and aligns with the UAE Net Zero by 2050 objectives. And while the use of energy recovery is from generated heat and converting it into usable energy is achieved in other industries, we hope that one day we can incorporate these technologies in the oil and gas industry.

Introduction

Global energy demand, especially within fossil-fuel-dominated industries, continues to exacerbate atmospheric emissions of pollutants and greenhouse gases (GHGs). Conventional waste gas flaring and incineration practices in the oil and gas sector contribute significantly to environmental degradation. The World Bank reported that global gas flaring reached its highest level since 2019, releasing substantial quantities of CO₂, methane, and soot into the atmosphere ([World Bank, 2024](#)). In the Middle East, particularly the UAE, associated gas flaring remains a challenge, due to the high H₂S content and the difficulty and cost required to treat, despite national efforts to reduce it, posing health risks—as recognized by the [IARC \(2002\)](#)—and hindering climate goals outlined in the Paris Agreement ([UN, 2015](#)) and national Net Zero commitments, such as the UAE's Strategic Initiative (MOCCAE, 2021). The increasing pressure from investors and the public on oil and gas companies to improve their ESG (Environmental, Social, and Governance) performance further underscores the urgent need for sustainable waste management solutions.

In traditional flaring incomplete combustion can lead to emissions of CO, unburnt hydrocarbons, and NO_x, all detrimental to health and the environment, since Flares are not 100% efficient, and some methane (un-combusted) is emitted during flaring. Also, flares cannot be performance tested to guarantee that they achieve the same efficiency as an incinerator or enclosed combustor. ("The differences between Flares, Incinerators and Enclosed Combustors ...") Studies suggest that the efficiency of a flare during windy conditions can be as low as 50%, while exhibiting a visible flame, as well as the fact that traditional flares cannot be used within 500 meters of residences. As industries pursue decarbonization, a persistent gap exists in sustainable and efficient waste gas treatment technologies that can integrate renewable energy and demonstrate robust performance under diverse operational conditions.

Incineration technologies have emerged as viable alternatives, as they demonstrated to have no smoke, No Odor and No visible flame, while offering complete combustion, so there is no un-combusted gas. Thus, it can combust heavier gases and eliminate odours (typically caused by odourises or mercaptans), and due to that it can be used within 500 meters of residences. ("The differences between Flares, Incinerators and Enclosed Combustors ...") It particularly emphasizes the treatment of complex petroleum waste gases, including Volatile Organic Compounds (VOCs) and hydrogen sulphide (H₂S), which are often overlooked in conventional carbon mitigation efforts. This study aims to establish Incinerators as scalable, environmentally responsible solutions aligned with regional and global environmental objectives and evolving ESG expectations.

Design and Component of the Waste Gas Incineration System

System Architecture

The core combustion unit is a vertically oriented, engineered to ensure stable axial flow and optimal temperature stratification. It's made from material specially selected for the harsh working environment of the demanding oil and gas industry.

The design promotes Natural air flow for continuous combustion that doesn't require any extra equipment like fans or pipes for air intake. crucial for the complete oxidation of hydrocarbons, VOCs, and H₂S within a short residence time. This chamber architecture ensures that the incinerators can work in the field for years with very little intervention or maintenance as some of the incinerators that were deployed in Canada 20 years ago are still operational today, saving huge amounts of emissions over the years

Control Architecture and Modularity. The incinerator's operation is monitored by a control system. Critical safety interlocks are integrated, including an over-temperature shutdown mechanism and other safety features, ensuring safe and reliable operation. A key design principle is modularity. The entire system is designed with skid-mounted, ISO-conformant modules, enabling rapid deployment within a 72-hour time frame, facilitating ease of transport, installation, maintenance, and scalability.

Material and Safety Consideration

Special alloys and coatings are used in the construction of the combustion and solar system due to their exceptional resilience to high temperature as well as corrosive gas components such as sulphur compounds and chlorinated hydrocarbons, aligning with recommendation from the US EPA 2023, these material choices ensure long term integrity and performance in demanding industrial environment, the system also incorporates multi-layer insulation of thermal efficiency and automated shut-off controls to comply with AER directive 060 ([Alberta Energy Regulator 2023](#)).

Methodology and CFD Modelling

Simulation Framework

Computational Fluid Dynamics (CFD) was used to optimize combustion performance and thermal uniformity by the OEM. ANSYS Fluent software was used for domain construction. Inlet gas compositions included 40% CH₄, 15% C₂H₆, 5% H₂S, and trace VOCs, with flow rates ranging from 5000-17000 cubic feet/hour, and ambient temperature of 40 degrees Celsius, and a pressure of 101.3 kPa.

Model Validation

Empirical validation was performed using a pilot incinerator deployed at a petroleum facility in Abu Dhabi. Flue gas analysis and thermal imaging confirmed a Destruction and Removal Efficiency (DRE) of >99.95% consistent with CFD prediction. Emissions were found to be within EU Directive 2000/76/EC and Alberta Directive 060 standards. Real time sensor data for thermal values and 5ppm for CO and NO_x, thus validating the simulation's reliability. Mesh independence was confirmed with less than 3% variation in key outputs when increasing the cell count from 5 million to 9 million, ensuring the computational results were not dependent on mesh resolution.

Parametric Study

A sensitivity analysis evaluated the impact of variations in solar input (2.8 kW/m²), gas flow rate, and ambient temperature on system performance. Stability decreased when parameters deviated by 10%, underscoring the necessity for real time control. NSGA-II optimization revealed that adaptive control could achieve 45% improvement in energy efficiency by balancing emission reduction, heat recovery, and fuel consumption ([Wang et al., 2024](#)).

Experimental Validation and Case Studies

Pilot Deployment in the UAE

A field pilot study was conducted in the United Arab Emirates (UAE) to evaluate the hybrid incinerator's performance under operational conditions. The deployment site was an oil and gas processing facility producing approximately 400,000 cubic feet of gas per day. Operated under the oversight of the client, following environmental guidelines supported by the Environmental Agency of Abu Dhabi (EAD). The incinerator was configured to process up to 16,700 cubic feet/hour of waste gas from crude oil production, including Methane CH₄, H₂S and BTEX (benzene, toluene, ethylbenzene, and xylene) compounds. The hybrid incinerator uses a solar powered ignition system to avoid cold flaring and ensure continuous flaring throughout flowing operations, compared to standard manual flare assisted systems. According to incineration monitoring criteria under EPA §264.343 and EU Directive 2000/76/EC the average combustion temperatures must consistently be maintained at 1050 degrees Celsius with nominal fluctuation. The installation of an active solar input stabilizer with field sensors would assist in maintaining the desired temperatures aiding in the Destruction and Removal Efficiency (DRE) of over 99.95% exceeding the [U.S EPA \(2023\)](#) baseline compliance threshold of 99.9%.

Comparative Case Study

In contrast to conventional burner system widely used in the oil and gas sector, the solar hybrid incinerator demonstrated in the current study offers substantial environmental and operational advantages. Traditional burners rely exclusively on fossil fuel combustion and operate at destruction EPA for hazardous waste incineration (U.S EPA, 2023). These systems also lack possible integrated heat recovery capabilities, resulting in substantial thermal energy losses and increased fuel costs. Emissions of (CO₂, 2000kg/ton, Nox 150ppm, CH₄ 80ppm and h₂S 70ppm) from traditional burners are markedly higher than those recorded for the hybrid system, which achieved reductions mentioned In Table 1. Moreover, traditional systems often rely on manual or periodic emission sampling, which limits real time responsiveness and prevents alignment with client expectation. In contrast, the hybrid system could possibly integrate automated real time gas monitoring ensuring both compliance and transparency in emission reporting.

Table 1—Emission Performance Comparison

System Type	CO ₂ Reduction%	Nox Reduction%	CH ₄ Reduction%	H ₂ S Reduction%
Traditional Burner	50	38	62	50
Hybrid Incinerator	67	60	75	77

Practical Applications in H₂S Treatment

In a truck loading facility in an EPF in Alberta, Canada; There was a need to eliminate H₂S coming from the associated sour gas from the produced light Oil, very similar to the region in the Middle East. Due to the very stringent regulations in the Area, the OEM installed a Thermocouple to measure the temperature in the combustion area, and if the temperature of the system drops below a certain level, this would activate a specialty burner, which introduces gas fuel to the system maintain a minimum temperature that insures 99% combustion meeting the AER (Alberta Energy Regulator).

Another Challenge in a different area arose when the Operator received complains about the naked flame and the odours resulting from traditional flaring, from the community and landowners. Even though the EPF was already in Operation the OEM deployed the mobile incinerators that were rigged up in ease replacing the old flare stake. The incinerator achieved 99.9% combustion and there were no more complains from the community regarding the foul odours.

Operational Cost Analysis

A detailed economic evaluation was conducted to assess capital expenditure (CAPEX), operatorial expenditure (OPEX), and return on investment (ROI). The initial installation cost of the hybrid incinerator is almost double of a traditional flare stake, inclusive of solar ignition, monitoring and control systems based on comparative costs in solar assisted waste to energy technologies (Wang et al 2024). Operating costs were reduced due to low requirements for maintenance and change of parts. Also, fuel cost savings compared to a traditional green burner, which used diesel for continuous burning of hydrocarbons in addition to water and other fluids from the well. An estimated future forecast of possible heat recovery and reuse could contribute to additional cost savings in compensated utility costs, based on models proposed under Canda's Clean Growth Program (Government of Canada, 2022).

Another cost aspect to the technology, is as it is a relatively new technology it has its limitations in terms of capacities. So, at the time being, through careful selection of types of wells and jobs to deploy the technology very affectively in, there still are some limitations in that aspect even though the footprint of the Incinerators is very small and has less restrictions in terms of placing during flaring, it won't be viable to deploy multiplies units.

Risk Assessment

Adhering to global guidance and as per [U.S. EPS \(2023\)](#) and [Alberta Energy Regulator \(2023\)](#) best practices, mitigation strategies included redundant ignition paths, real time flow sensors, and automated heat exchanger backflush. The possibility of these risks occurring is low with thorough maintenance, good control systems, and robust risk mitigation measures in place. However, as any new technology, local operators and service providers are still lacking enough experience and exposure with the knowledge to be able to develop detailed HAZOP, HAZID, JSAs etc. as we approach closer to the UAE's initiative for Net Zero 2050 (MOCCA, 2021).

Regulatory and Environment Compliance

Regional and Technical Standards

The system consistently meets key regulatory standards U.S. CFR §264.343. Destruction and Removal Efficiency (DRE) and emissions was within limits. EU Directive 2000/76/EC compliance was also achieved. The predicted CO₂ levels (below 10 ppm), unburned hydrocarbons, and NO_x concentration (30-36 ppm) align with international emission limits, demonstrating strong adherence to national and international air quality.

ESG Alignment and Global Regulatory Framework

Globally, waste incineration practices are governed by strict framework including the Basel Convention, which regulates the transboundary movement and disposal of hazardous waste. Hybrid systems such as the one evaluated in this study can be aligned with Basel guidelines by ensuring high temperature destruction of organics and implementing real time emissions monitoring. The [IARC \(2002\)](#) also emphasizes the health risks of particulate matter and combustion byproducts, validating the importance of clean burning, high efficiency incineration systems in both development and emerging economies.

The Paris Agreement's Article 4 encourages lower GHGs from industrial sectors. By significantly reducing emissions per ton of waste, this system directly strengthens national contributions to climate action ([UN, 2015](#)). In the UAE, it directly supports the Net Zero 2050 Strategic Initiative and enhances GHG inventory accuracy under MOCCA (2021). It also advances UN Sustainable recovery and substantial emissions reduction ([UN, 2015](#)). The integration of heat recovery and solar preheating helps petroleum operators align with the UAE's Climate Responsible Companies Pledge and sectoral decarbonization roadmaps.

From an ESG (Environmental, Social, and Governance) perspective, the system is exceptionally well suited to meet increasing investor and regulatory expectations. The inclusion of automated emission monitoring systems enables transparent reporting, aligning with framework like the TCFD (Task Force on Climate Related Financial Disclosures) and GRI (Global Reporting Initiative) metrics for environmental impact. Adoption of this system can significantly support oil and gas operators in improving their ESG ratings. Its high efficiency and low start up emissions qualify it for funding from various green initiatives, such as Canada's Clean Growth Program (2022).

Enhancing Innovation Design for Waste Gas Incineration

The theoretical enhancing framework for the waste gas incineration system is to integrate high temperature combustion chamber with advanced energy input and recovery mechanism, this design is to efficiently process diverse waste gas streams including various hydrocarbon, VOC & H₂S by introducing them into combustion zones throughout preheat pathway.

Solar Assisted Ignition Module

The key of enhancing the system with solar assisted ignition module is to harness the concentrated solar thermal energy, typically via parabolic trough arrays, to preheat the incoming hydrocarbon flow, to target an optimal temperature of 300-400 degrees Celsius at the burner inlet, as it will be significant reduction auxiliary energy required for ignition and substantial decrease in the cold start duration, this will leverage external renewable energy to improve the thermodynamics efficiency of the heating element, powered by thermoelectric generator, to ensure continuous operation readiness independent of solar insulation level.

Integrated Heat Recovery. To maximize energy efficiency and sustainability is to contribute to the system's overall, three distinct subsystems are integrated for thermal energy recovery, achieving a combined efficiency exceeding 72%.

- **Shell and Tube Exchange:** This primary heat recovery process unit serves to preheat process water or air to a temperature range of 70-80 degrees Celsius suitable for various industrial or domestic applications.
- **Thermoelectric Generator (TEG) Array:** An array of Thermoelectric Generators is strategically placed to convert a portion of the waste heat directly into DC electrical power, with an expected output of up to 420W, to contribute self-sufficiency.
- **Phase Change Material (PCM) storage:** A crucial component, to provide a thermal buffering to extend the availability of recovered low grade heat, by utilizing material with melting point approx. 140 degrees Celsius, this system will capture and release the latent heat during phase transition.

Heat Recovery System. Post-combustion gases are routed through a heat exchanger to extract residual energy, which can be utilized for preheating feed gases, water desalination, or space heating applications. This energy recovery process significantly enhances system efficiency and reduces the external energy input required per unit of waste gas destroyed. The economic viability of this configuration for large scale facilitates is supported by techno-economic analysis ([Wang et al., 2024](#)).

Conclusion

The findings of this study demonstrates that solar assisted incinerators offer a practical and highly effective path toward environmentally responsible waste gas management in the petroleum sector. The assessment conducted on this system indicates a significant reduction in overall carbon footprint compared to traditional flaring of. waste gas treatment methods, reinforcing its sustainability capability. By synergizing the integration of solar thermal systems, advanced combustion, and comprehensive energy recovery, such systems provide a benefit of profound emission reduction and superior energy efficiency.

Future work should explore the development of digital twin applications for enhancing control, which would maximize up time and minimize operational expenditure (OPEX) through predictive maintenance systems, building greater resilience into remote or offshore facility applications. Expansion into offshore platforms presents unique opportunities, despite space constraints and complex safety regulations, due to their significant flaring challenges. Integration with carbon capture modules can further reduce residual carbon dioxide emissions, supporting efforts for other industrial sectors. Funding initiatives like the Clean Growth Program ([Government of Canada, 2022](#)) and implementation frameworks under the UN Sustainable Development Goals ([UN, 2015](#)) can cause deployment at scale. With proper stakeholder engagement and infrastructural planning, hybrid incineration technology has the transformative potential to reshape waste gas management practices across the global petroleum industry. Collaboration with EPC (Engineering, Procurement and Construction) contractors and technology licenses will be essential to regulate modular incinerator packages for broader global deployment. The integration with AI driven predictive maintenance systems can further enhance up time, minimize OPEX, and build resilience into remote or offshore facility applications, aligning with resilience strategies outlined by the [World Bank \(2024\)](#).

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